

BHOOMIKA GOEL

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SUMMARY

AI/ML Engineer specializing in agentic LLM systems and RAG pipelines, with project experience across LangChain, LangGraph, PyTorch, FastAPI, and SHAP-based explainability. Architected a 6-agent autonomous research system, a hallucination-detecting RAG pipeline, and a privacy-preserving federated learning framework — validated through rigorous statistical testing.

PROJECTS

ARIA — Autonomous Research Intelligence System

2026

Python, LangGraph, Gemini, Groq, ChromaDB, Sentence Transformers

- Architected a **6-agent autonomous system** (Planner, Research, Summarizer, Synthesis, Critic, Formatter) enabling task decomposition, **cross-document reasoning**, **critique-driven refinement**, and end-to-end **research report generation**.
- Designed stateful semantic memory with ChromaDB and vector embeddings, supporting similarity-based retrieval, adaptive knowledge reuse, and reduced redundant LLM inference across sessions.
- Implemented multi-agent orchestration with retry/fallback routing, semantic ranking, caching, and observable stateful workflows for reliable research automation.

TriMind AI — Advanced RAG Pipeline with Evaluation Layer

2026

Python, LangChain, Pinecone, Gemini, FastAPI

- Built a modular RAG pipeline with query rewriting, MMR-based retrieval, and conversational memory for context-aware, source-grounded responses across large document sets.
- Designed an evaluation-driven system with grounding and confidence scoring and hallucination detection to improve response reliability and transparency.
- Developed a FastAPI backend integrating LangChain, Pinecone, Gemini API, caching, and structured logging for production-style AI querying.

FL-XAI — Federated Learning for Parkinson's Disease Prediction

2026

Python, Scikit-learn, SHAP, Pandas, NumPy

- Built a **privacy-preserving multi-round federated learning framework** across **5 heterogeneous clients** for UPDRS prediction under non-IID conditions, achieved **RMSE 4.51** and **R² 0.8162**.
- Developed adaptive aggregation (RMSE-based weighting, client-size normalization, ensemble fusion) and integrated SHAP explainability with Kendall Tau consistency and Bootstrap CI + Wilcoxon validation.

Sunspot Forecasting — Ensemble Deep Learning

2025

Python, PyTorch, TensorFlow, PyTorch Lightning

- Benchmarked **9 deep learning architectures** on **250+ years** of SILSO data; custom CNN+LSTM/GRU hybrids consistently outperformed baseline models across RMSE, MAE, and R².

TECHNICAL SKILLS

Generative AI & Agentic Systems: LLMs, RAG, LangChain, LangGraph, Hugging Face, Pinecone, ChromaDB, Prompt Engineering

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, PyTorch Lightning, CNN, LSTM, GRU, Ensemble Learning, Time Series Forecasting

Federated Learning & XAI: Federated Learning, Explainable AI (SHAP), Adaptive Aggregation, Statistical Validation

Backend & APIs: FastAPI, Streamlit, Docker, REST APIs

Programming: Python, C++

Data & Development: NumPy, Pandas, Matplotlib, Git, GitHub

PROFESSIONAL EXPERIENCE

AI/ML Virtual Intern — EduSkills, Google for Developers, AICTE

Jan 2025 – Mar 2025

- Implemented classification and prediction models using PyTorch/TensorFlow with preprocessing and feature engineering; built Streamlit demos for evaluation visualisation. Grade: A

Google Cloud GenAI Virtual Intern — SmartBridge & SmartInternz (AICTE)

Sep 2024 – Oct 2024

- Built Generative AI applications on Google Cloud using Streamlit; designed end-to-end pipelines with prompt engineering for text generation and automation workflows.

ServiceNow Academic India Cohort — ServiceNow

Mar 2025 – Aug 2025

- Completed advanced training in **Agentic AI, ServiceNow Administration**, workflow automation, **Automated Test Framework (ATF)**, strengthening enterprise AI and platform automation skills.

EDUCATION

B.Tech in Computer Science & Engineering (AI Specialization) — CGPA: 9.32/10

Oct 2023 – Present

Dr. A.P.J. Abdul Kalam Technical University, MIET Meerut

DAV Public School (CBSE) — XII: 93.8% | X: 95.8%

2023 / 2021

CERTIFICATIONS

- **Oracle Certified Foundations Associate – Agentic AI** — Oracle University | Jul 2026 – Jul 2028
- **AWS AI Practitioner Challenge** — AWS × Udacity | Jun 2026
- **Digital Application Fundamentals – STEM (Gold, 94.78%)** — NASSCOM, Govt. of India | Mar 2025
- **Introduction to Agentic AI** — ServiceNow | Oct 2025
- **Junior Cybersecurity Analyst Career Path & Data Analytics Essentials** — Cisco | Jan 2025

LEADERSHIP & ACHIEVEMENTS

- Secretary, Microsoft Azure Community (MIET); Graphic Designer Head, Society of Electronics Engineers (SEE)
- Ranked among the **Top 15 B.Tech CSE students** during the first and second academic years at MIET.